Practice Task 3

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1 fx IF

IF

What does it do?

It checks whether a condition is true or not and on the basis of that returns a value

Syntax

=if(Logical_Test, [value_if_true], [value_if_false])

Logical_Test

This will be the condition which you want to check, so greater than, less than or equal to signs can be used for numbers. When you want to evaluate a condition based on text value, then use double quotes (") with equal to signs (Row 27 for example)

Value_if_True

If the logical test holds, then excel will return this value

Value_if_False

If the logical test does not hold, then excel will return this value

Example

Using IF formula to identify if a student has passed an exam or not

Student ID	Marks	Pass or Fail	Formula
S001	90	Pass	=IF(B22>=40,"Pass","Fail")
S002	70	Pass	=IF(B23>=40,"Pass","Fail")
S003	20	Fail	=IF(B24>=40,"Pass","Fail")
S004	41	Pass	=IF(B25>=40,"Pass","Fail")

Note

If instead of marks, we had Grades A, B, C i.e. text values then the above formulas would have changed by

IF(B21= 'A',"Pass","Fail")

Exercise

Write a formula using IF. Salary for an AVP will be $2,000 and for an analyst will be $1,000

Emp ID	Designation	Salary	Formula
E001	AVP	$2,000	=IF(B34="AVP",2000,1000)
E002	Analyst	$1,000	=IF(B35="AVP",2000,1000)
E003	Analyst	$1,000	=IF(B36="AVP",2000,1000)
E004	Analyst	$1,000	=IF(B37="AVP",2000,1000)

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1 AND

AND

What does it do?

Checks if all arguments are true and returns TRUE value else returns FALSE value. You need to have minimum 2 conditions to check. AND function is used with IF condition a lot

Syntax

=AND(Logical 1, Logical 2, Logical 3, and so on..)

Logical 1

First condition which needs to be checked

Logical 2

Second condition which needs to be checked

Example 1

Testing condition 1 and condition 2 to check the numbers are greater than 0 using AND

Condition 1	Condition 2	Result	Formula
1	2	TRUE	=AND(A19>0,B19>0)
1	-2	FALSE	=AND(A20>0,B20>0)

Example 2

Use IF with AND function to check if student has passed or not. If the student has failed in 1 of the 2 subjects, then the student will be considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	PASS	=IF(AND(B26="PASS",C26="PASS"),"PASS","FAIL")
S002	PASS	FAIL	FAIL	=IF(AND(B27="PASS",C27="PASS"),"PASS","FAIL")
S003	FAIL	PASS	FAIL	=IF(AND(B28="PASS",C28="PASS"),"PASS","FAIL")
S004	PASS	PASS	PASS	=IF(AND(B29="PASS",C29="PASS"),"PASS","FAIL")

Exercise

Write a formula using AND with IF to find if an employee should get a "GOOD" increment or not. Sales made by the employee should be more than 50 and he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)	Formula
E001	AVP	51	YES	YES	=IF(AND(C36>50, D36="YES"), "YES", "NO")
E002	Analyst	100	NO	YES	=IF(AND(C37>50, D36="YES"), "YES", "NO")
E003	Analyst	49	YES	NO	=IF(AND(C38>50, D36="YES"), "YES", "NO")
E004	Analyst	75	YES	YES	=IF(AND(C39>50, D36="YES"), "YES", "NO")

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 Nested IF's

Nested IF's

What does it do?

Nested IF conditions help in checking multiple conditions together by using IF conditions within IF condition.

Syntax

=if(Logical_Test, [value_if_true], if(Logical_Test, [value_if_true], [value_if_false]))

Logical_Test

This will be the condition which you want to check, so greater than, less than or equal to signs can be used for numbers. When you want to evaluate a condition based on text value, then use double quotes ("") with equal to signs

Value_if_True

If the logical test holds, then excel will return this value

Value_if_False

If the logical test does not hold, then excel will return this value

Nested IF

From Excel 2007 onwards, we can use 64 IF conditions in one formula whereas Excel 2003 allows for ony 7 IF conditions

Example

Use Nested IF to put the grade a student gets in an exam in Table B. Score Ranges for different grades are given in Table A.

TABLE A	
Score	Grade
> 70	A
41-70	B
< 40	C

TABLE B			
Student ID	Marks	Grade	Formula
S001	90	A	=IF(E26< 41,"C",IF(E26< 71,"B","A"))
S002	70	B	=IF(E27< 41,"C",IF(E27< 71,"B","A"))
S003	20	C	=IF(E28< 41,"C",IF(E28< 71,"B","A"))
S004	41	B	=IF(E29< 41,"C",IF(E29< 71,"B","A"))

Exercise

Write a formula using Nested IF conditions to check what level increment an employee should get and then convert the amount into dollars using Format

TABLE A	
Performance	Increment
A	$1,000
B	$500
C	$250

TABLE B				
Emp ID	Designation	Performance	Result	Formula
E001	AVP	A	1000	=IF(F36="A", 1000, IF(F36="B", 500, 250))
E002	Analyst	A	1000	=IF(F37="A", 1000, IF(F37="B", 500, 250))
E003	Analyst	B	500	=IF(F38="A", 1000, IF(F38="B", 500, 250))
E004	Analyst	C	250	=IF(F39="A", 1000, IF(F39="B", 500, 250))

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1 OR

OR

What does it do?

Checks if any of the conditions are true or not. It returns TRUE value even if one of the conditions holds else returns FALSE value. Basically, for a FALSE to appear, no conditions should hold. You need to have minimum 2 conditions to check. AND function is used with IF condition a lot

Syntax

=OR(Logical 1, Logical 2, Logical 3, and so on..)

Logical 1

First condition which needs to be checked

Logical 2

Second condition which needs to be checked

Example 1

Testing condition 1 and condition 2 to check the numbers are greater than 0 using OR

Condition 1	Condition 2	Result	Formula
1	2	TRUE	=OR(A20>0,B20>0)
1	-2	TRUE	=OR(A21>0,B21>0)
-1	-2	FALSE	=OR(A22>0,B22>0)

Example 2

Use IF with OR function to check if student has passed or not. If the student has failed in 1 of the 2 subjects, then the student will be considered as Failed

Student ID	Subject 1	Subject 2	Overall Result	Formula
S001	PASS	PASS	PASS	=IF(OR(B28="FAIL",C28="FAIL"),"FAIL","PASS")
S002	PASS	FAIL	FAIL	=IF(OR(B29="FAIL",C29="FAIL"),"FAIL","PASS")
S003	FAIL	PASS	FAIL	=IF(OR(B30="FAIL",C30="FAIL"),"FAIL","PASS")
S004	PASS	PASS	PASS	=IF(OR(B31="FAIL",C31="FAIL"),"FAIL","PASS")

Exercise

Write a formula using OR with IF to find if an employee should get a "GOOD" increment or not. Either Sales made by the employee should be more than 50 or he should have adhered to all HR policies

Emp ID	Designation	Sales	Adhered to HR Policies	GOOD Increment (Yes or NO)	Formula
E001	AVP	51	YES	YES	=IF(OR(C38>50, D38="YES"), "YES", "NO")
E002	Analyst	49	NO	NO	=IF(OR(C39>50, D39="YES"), "YES", "NO")
E003	Analyst	49	YES	YES	=IF(OR(C40>50, D40="YES"), "YES", "NO")
E004	Analyst	75	YES	YES	=IF(OR(C41>50, D41="YES"), "YES", "NO")

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 IFERROR

IFERROR

What does it do?

Sometimes when we use formula in cells, we might get an error because of the cell value, IFERROR allows us to remove such errors with values we want

Syntax

=IFERROR(Value,[Value_if_error])

Logical 1

First cell value which is checked if there is an error

Value_if_error

If in the cell there is an error, that error value will be replaced by this value

Example 1

Divide Col A by Col B and show the results without using IFERROR and with using IFERROR

Col A	Col B	Without IFERROR &Formula		With IFERROR &Formula	
1	2	0.5	=A19/B19	0.5	=IFERROR(A19/B19,999)
1	0	#DIV/0!	=A20/B20	999	=IFERROR(A20/B20,999)

Note

In the above example, on dividing 1 by 0, we get an error, without using IFERROR we get #DIV/0! i.e. an error. So using IFERROR, we can give values we want. You can also enter text which can let the user know that there is some problem with the data.

Example 2

Divide Col A by Col B and show the results without using IFERROR and with using IFERROR

Col A	Col B	Without IFERROR &Formula		With IFERROR &Formula	
1	2	2	=A29*B29	2	=IFERROR(A29*B29,"DATA IS WRONG")
1	1	1	=A30*B30	1	=IFERROR(A30*B30,"DATA IS WRONG")
1	a	#VALUE!	=A31*B31	DATA IS WRONG	=IFERROR(A31*B31,"DATA IS WRONG")
5	a	#VALUE!	=A32*B32	DATA IS WRONG	=IFERROR(A32*B32,"DATA IS WRONG")

The red color highlight is for representation here - color cannot be changed using IFERROR

Exercise

Write a formula using IFERROR. Find the monthly salary of each employee using data in Column C and if an error appears, then change it to ERROR IN SALARY. Change the format to £ (Pounds) and no decimal places

Emp ID	Designation	Annual Salary	Monthly Salary	Formula
E001	AVP	£80,000	£6,667	=IFERROR(C39/12,"ERROR IN SALARY")
E002	Analyst	£50,000	£4,167	=IFERROR(C40/12,"ERROR IN SALARY")
E003	Analyst	abcd	ERROR IN SALARY	=IFERROR(C41/39, "ERROR IN SALARY")
E004	Analyst	£40,000	£3,333	=IFERROR(C42/12,"ERROR IN SALARY")

EXACT

What does it do?

It checks whether 2 values (texts, numbers) are exactly same or not. If they are same, then it will return TRUE else it will return FALSE. This function is case-sensitive

Syntax

=EXACT(Text1, Text2)

Text1

First string which you want to check

Text2

Second string which you want to test against

Example 1

Testing condition 1 and condition 2 to check the numbers are greater than 0 using AND

Col A	Col B	Result	Formula
EXCEL	EXCEL	TRUE	=EXACT(A19,B19)
EXCEL	Excel	FALSE	=EXACT(A20,B20)

Note

In the above example, Row 20 result is FALSE because the EXACT function is case-sensitive. The first value is in upper capse (EXCEL) whereas the second one (Excel) is a combination of upper and lower capse

Exercise

Write a formula using EXACT to check if the values in Col A and Col B are same or not

Col A	Col B	Results	Formula
TEST	Test	**FALSE**	=EXACT(A29,B29)
TEST	TEST	**TRUE**	=EXACT(A30,B30)
1234	1234	**TRUE**	=EXACT(A31,B31)
1.1	1.101	**FALSE**	=EXACT(A32,B32)

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 | fx TEXT

TEXT

What does it do?

It converst a numeric value into text based on the format given by the user

Syntax

=TEXT(Value, Format_text)

Value

Value for which we need to change the format

Format_text

Format specified by the user

Example 1

Using different text formats

Col A	Result	Formula
3/27/2017	27/03/2017	=TEXT(A18, "dd/mm/yyyy")
42821	27/3/2017	=TEXT(A19, "dd/m/yyyy")
42821	Mar 27 2017	=TEXT(A20, "mmm dd yyyy")
3/27/2017	Mar-17	=TEXT(A21,"mmm-yy")
14.141414	14.14	=TEXT(A22, "0.00")
14598.12	$14,598.12	=TEXT(A23, "$#,##0.00")
14598.12	£14,598.12	=TEXT(A24, "£#,##0.00")
14:00	14:00	=TEXT(A25, "hh:mm")

Using Text function though the results may appear in the same way, but there format has changed to text.

Note:

Col A Row 19 & 20 - Excel stores dates in numbers, so that is why when we use Text with Date formats, Excel converts those numbers dates

Exercise

Write a formula using EXACT to check if the values in Col A and Col B are same or not

Value	Convert To	Results	Fromula
5/21/2017	5/21/2017	**21/05/2017**	=TEXT(A34, "dd/mm/yyyy")
42876	5/21/2017	**21/05/2017**	=TEXT(A35, "dd/mm/yyyy")
12:00	12:00	**0.50**	=TEXT(A36, "0.00")
12345.78	£12,345.78	**£12,345.78**	=TEXT(A37,"£##,###.00")
12345.78	£12,346	£12,346	=TEXT(A38,"£##,###")

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:C1 CONCATENATE

CONCATENATE

What does it do?

This function joins several (more than 1) text string into 1 string

Syntax

=CONCATENATE(String1, String2, and so on..)

OR

=String1&String2& so on..

String 1

First string which you want to concatenate

String 2

Second string which you want to join to the first string

Example

Join the 3 strings together using CONCATENATE and & (ampersand) operator

String1	String2	String3	Result	Formula
S001	90	Pass	**S00190Pass**	=CONCATENATE(A20,B20,C20)
S002	70	Pass	**S00270Pass**	=CONCATENATE(A21,B21,C21)
S003	20	Fail	**S00320Fail**	=A22&B22&C22
S004	41	Pass	**S00441Pass**	=A23&B23&C23
S005	39	Fail	**S005 39 Fail**	=CONCATENATE(A24," ",B24," ",C24)
S006	99	Pass	**S006 99 Pass**	=A25&" "&B25&" "&C25
S007	11	Fail	**S007;11;Fail**	=CONCATENATE(A26,";",B26,";",C26)

In these 3 examples, we have added spaces or semi-colons between the strings to make the result readable

Note

In order to add spaces, other special characters between strings, you would need to use double quotes "" and declare the value between them as done in the examples above

Exercise

Join the 3 strings together using CONCATENATE and & (ampersand) operator to get the results as mentioned in the result column

String1	String2	String3	Result	USE	Formula
E001	AVP	800	**E001-AVP-800**	CONCATENATE	**=CONCATENATE(A35, B35, C35)**
E002	Analyst	500	**E002Analyst500**	CONCATENATE	**=CONCATENATE(A36, B36, C36)**
E003	Analyst	100	**E003/Analyst-100**	CONCATENATE	**=CONCATENATE(A37, B37, C37)**
E004	Analyst	400	**E004:Analyst:400**	&	**=CONCATENATE(A38, B38, C38)**
E005	AVP	1000	**E005 AVP 1000**	&	**=CONCATENATE(A39, B39, C39)**
E006	Analyst	250	**E006 Analyst 250**	&	**=CONCATENATE(A40, B40, C40)**
E007	Analyst	500	**E007;Analyst*500**	&	**=CONCATENATE(A41, B41, C41)**

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 TRIM

TRIM

What does it do?

The function removes spaces all before and following the words i.e. leading and trailing spaces - and also removes extra spaces between words but does not remove the single space between words. TRIM function is used to clean data.

Syntax

=TRIM(Text)

Text

First string which you want to concatenate

Example

Use TRIM to remove spaces

String1	Result	Formula
S001 A	**S001 A**	=TRIM(A16)
S001 A	**S001 A**	=TRIM(A17)
S002	**S002**	=TRIM(A18)
S003	**S003**	=TRIM(A19)

Exercise

Remove the extra spaces in the strings and then join the 3 strings together using CONCATENATE and & (ampersand) operator to get the results as mentioned in the result column. Do so in one formula itself

String1	String2	String3	Result	USE	Formula
E001	AVP	800	**E001-AVP-800**	CONCATENATE	**E001**
E002	Analyst	500	**E002Analyst500**	CONCATENATE	**E002**
E003	Ana lyst	100	**E003/Analyst-100**	&	**E003**
E004	Analyst	400	**E004:Analyst:400**	&	**E004**

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:C1 SUBSTITUTE

SUBSTITUTE

What does it do?

This function is used to replace a part of text in a string with something else. SUBSTITUTE function is used to clean data. This function is case-sensitive.

Syntax

=SUBSTITUTE(Text, Old_Text, New_Text, [instance_number])

Text

String in which you want to modify text

Old_Text

The old text which you want to replace

New Text

The new text with which you want to modify your original text with

Instance_Number

The instance of old text which you want to replace with new text. It is optional but if not provided it will make changes to all the occurrences of old text with new text. Look at the examples below for better understanding of this

Example

Join the 3 strings together using CONCATENATE and & (ampersand) operator

	String	Instances	Result	Formula
26	Hut		Hub	=SUBSTITUTE(A26,"t","b")
27	bumble	1	Tumble	=SUBSTITUTE(A27,"b","T",1)
28	the car is in the driveway		car is in driveway	=SUBSTITUTE(A28,"the"," ")
29	the car is in the driveway	2	the car is in a driveway	=SUBSTITUTE(A29,"the","a",2)
30	The car is in the driveway		car is in the driveway	=SUBSTITUTE(A30,"The"," ")

Note

If instances are not mentioned, then excel will replace all the occurrences as can be seen on Row 28. And the function is case-sensitive which can be seen on Row 30

Exercise

Use SUBSTITUTE function to make changes to some of the strings below to get the results mentioned in the 'Result' column

	String	Result	Formula	Formula
39	The dog is in the lawn	dog is in the lawn	**dog is in the lawn**	=SUBSTITUTE(A39, "The ", "")
40	the dog is in the lawn	Dog is in lawn	**dog is in lawn**	=SUBSTITUTE(A40, "the", "")
41	Dog	Fog	**Fog**	=SUBSTITUTE(A41, "D", "F")
42	Curious Cat	Furious Cat	**Furious Cat**	=SUBSTITUTE(A42, "C", "F", 1)

Menus 100% View only

A1:B1 | VALUE

VALUE

What does it do?

The function is used to convert text value into numeric value

Syntax

=VALUE(Text)

Text

String which you want to convert to number

Example

Convert the values into numbers

Text	Result	Formula
$12,345	**12345**	=VALUE(A15)
3/27/2017	**42821**	=VALUE(A16)
12/9/2014	**41982**	=VALUE(A17)

Excel stores dates as numbers, hence the function changes the dates to numbers

Exercise

Convert the values in Text column into numbers

Text	Result	Formula
$123.17	**123**	**=VALUE(A22)**
12/1/2017	**43070**	**=VALUE(A23)**
1/12/2017	**42747**	**=VALUE(24)**

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 fx UPPER

UPPER

What does it do?

This function converts the entire text string into uppercase letters

Syntax

=UPPER(Text)

Text

String for which you want to convert all letters into UPCASE

Example

Convert the values in Text column into UPPERCASE

Text	Result	Formula
Laptop	LAPTOP	=UPPER(A15)
excel	EXCEL	=UPPER(A16)
superstore is closed now	SUPERSTORE IS CLOSED NOW	=UPPER(A17)

Exercise 1

Convert the values in Text column into UPPERCASE

Text	Formula
good	**=UPPER(A22)**
how are you?	**=UPPER(A23)**

Exercise 2

Use UPPER to change Col A and then use EXACT function to match the 2 strings to see if they are same or not

Col A	Col B	Result
match	MATCH	MATCH
match	Match	MATCH
correct	CORECT	CORRECT

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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Menus 100% View only

A1:B1 fx LOWER

LOWER

What does it do?

This function converts the entire text string into lowercase letters

Syntax

=LOWER(Text)

Text

String for which you want to convert all letters into lowercase

Example

Convert the values in Text column into lowercase

Text	Result	Formula
Laptop	laptop	=LOWER(A15)
EXCEL	excel	=LOWER(A16)
SUPERSTORE IS CLOSED NOW	superstore is closed now	=LOWER(A17)

Exercise 1

Convert the values in Text column into lowercase

Text	Formula
GOOD	**=LOWER(A22)**
HOW are you?	**=LOWER(A23)**

Exercise 2

Use LOWER to change Col A and then use EXACT function to match the 2 strings to see if they are same or not

Col A	Col B	Result
MATCH	match	match
MATCH	mATCH	match
CORECT	correct	corect

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

File Edit View Insert Format Data Tools Help

Menus 100% View only

A1:B1 LEFT

LEFT

What does it do?

This function returns the number of specified characters from the start of the string. This function is used for data manipulation. A way to remember is that you read from left to right - so that is why from the start of the string

Syntax

=LEFT(Text, [num_chars])

Text

String from which you want to find letters

num_chars

Number of characters you want to read from the left

Example

Use LEFT function on the following strings

Text	Result	Formula
Laptop	Lapto	=LEFT(A19,5)
EXCEL IS WOW	EXCEL IS	=LEFT(A20,8)
SUPERSTORE IS CLOSED ;	SUPERSTORE IS C	=LEFT(A21,15)
90	9	=LEFT(A22,1)
SUPERSTORE IS CLOSED ;	SUPERSTORE IS C	=LEFT(A23,LEN(A23)-7)

Exercise 1

Use LEFT function to get the results in the 2nd column

Text	Result	Formula
Phone Number is +(0)1234	Phone Number is +(0	=LEFT(A28, 19)
This is used for data cleaning	This is used for data cle	=LEFT(A29, 25)

Exercise 2

Use LEFT and LEN function together to only remove last 5 characters of the string

Text	Result	Formula
Phone Number is +(0)1234	Phone Number is +(0	=LEFT(A34, LEN(A34)-5)
This is used for data cleaning	This is used for data cle	**=LEFT(A35, LEN(A35)-5)**

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

File Edit View Insert Format Data Tools Help

Menus 100% View only

A1:B1 fx LEN

LEN

What does it do?

This function returns the number of characters in a string (text or numeric value). It also counts the spaces and special characte
This function is used for data manipulation along with LEFT, RIGHT functions

Syntax

=LEN(Text)

Text

String for which you want to find the length

Example

Convert the values in Text column into lowercase

Text	Result	Formula
Laptop	6	=LEN(A16)
EXCEL	6	=LEN(A17)
SUPERSTORE IS CLOSED ;	22	=LEN(A18)
90	2	=LEN(A19)

Exercise 1

Find the length of the followig strings

Text	Formula
Phone Number is +(0)1234	=LEN(A24)
This is used for data cleaning	=LEN(A25)

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

File Edit View Insert Format Data Tools Help

Menus 100% View only

A1:B1 fx RIGHT

RIGHT

What does it do?

This function returns the number of specified characters from the end of the string. This function is used for data manipulation. A way to remember is that you read from left to right - so that is why from the end of the string

Syntax

=RIGHT(Text, [num_chars])

Text

String from which you want to find letters

num_chars

Number of characters you want to read from the left

Example

Use RIGHT function on the following strings

Text	Result	Formula
Laptop	top	=RIGHT(A19,3)
EXCEL IS WOW	WOW	=RIGHT(A20,4)
SUPERSTORE IS CLOSED ;	OSED ;	=RIGHT(A21,6)
90	0	=RIGHT(A22,1)
SUPERSTORE IS CLOSED ;	OSED ;	=RIGHT(A23,LEN(A23)-16)

Exercise 1

Use RIGHT function to get the results in the 2nd column

Text	Result	Formula
Phone Number is +(0)1234	is +(0)1234	=RIGHT(A28,11)
This is used for data cleaning	data cleaning	=RIGHT(A29,13)

Exercise 2

Use RIGHT and LEN function together to only remove first 5 characters of the string

Text	Result	Formula
Phone Number is +(0)1234	Number is +(0)1234	=RIGHT(A34,LEN(A34)-11)
This is used for data cleaning	is used for data cleaning	**=RIGHT(A35,LEN(A35)-13)**

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

File Edit View Insert Format Data Tools Help

Menus 100% View only

A1:B1 fx MID

MID

What does it do?

This function returns the number of specified characters from the left to right but, you can change the start of the string thus making MID function different from LEFT function.

Syntax

=MID(Text, [starting_position], [num_chars])

Text

String from which you want to find letters

starting_position

Starting position of the string

num_chars

Number of characters you want to read from the starting of the string

Example

Use RIGHT function on the following strings

Text	Result	Formula
Laptop restarted	top re	=MID(A22,4,6)
EXCEL IS WOW	XCEL IS W	=MID(A23,2,9)
SUPERSTORE IS CLOSED ;	E IS CLOS	=MID(A24,10,9)
90 EMPLOYEES	0 EMPL	=MID(A25,2,6)

Exercise

Use MID function to get the results in the 2nd column

Text	Result	Formula
Phone Number is +(0)1234	is +(0)1	=MID(A30,5,7)
This is used for data cleaning	for data clean	=MID(A31,9,14)
How have you been?	you	=MID(A32,9,3)
Any questions?	quest	=MID(A33,6,5)

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 fx DATE

DATE

What does it do?

This function returns the date when days, month, year are stored in different cells. If the cell in which the function is being called is of Number format, then the function will return a number - excel stores dates as numbers

Syntax

=DATE(Year,Month,Day)

Year

Year of the date. Try to supply the entire 4 digits

Month

Month of the date in numeric format like for January it would be 1, for February it would be 2

Day

Day of the date - it needs to be in numeric format like 1, 2, 3 ...and till 31

Example

Use DATE function to calculate dates

Year	Month	Day	Result	Formula
2017	12	31	12/31/2017	=DATE(A22,B22,C22)
2016	11	30	11/30/2016	=DATE(A23,B23,C23)
2015	3	1	3/1/2015	=DATE(A24,B24,C24)
2014	8	1	8/1/2014	=DATE(A25,B25,C25)

Exercise 1

Use DATE function to calculate dates and then format the result into Date format

Year	Month	Day	Formula
2012	7	31	7/31/2012
2011	1	30	1/30/2011
2010	2	1	2/1/2010
2000	5	25	5/25/2000

Exercise 2

Use LEFT, MID and RIGHT functions to extract day, month and year from Col A and then use DATE function to calculate dates

Col A	Year	Month	Day	Date	Formula
21122016	2016	12	21	12/21/2016	=DATE(B38, C38, D38)
13122000	2000	12	13	12/13/2000	=DATE(B39, C39, D39)
15011999	1999	01	15	1/15/1999	=DATE(B40, C40, D40)
10102010	2010	10	10	10/10/2010	=DATE(B41, C41, D41)

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 | fx DATEVALUE

DATEVALUE

What does it do?

This function converts a date which has been entered as text to date value but in number format

Syntax

=DATEVALUE(DATE_TEXT)

DATE_TEXT

Date which needs to be converted from text to date format

Example

Use DATE function to calculate dates

Text Date	Result	Formula
05/05/2017	42860	=DATEVALUE(A15)
02/12/2018	43143	=DATEVALUE(A16)
01/03/2015	42007	=DATEVALUE(A17)
02/08/2014	41678	=DATEVALUE(A18)

Exercise

Use DATEVALUE function to calculate dates and then change the format of the result into Date format

Text Date	Result	Formula
31/12/2015	42369	=DATEVALUE("12/31/2015")
3/01/2010	40238	=DATEVALUE("03/01/2010")
11/12/2015	42320	=DATEVALUE("11/12/2015")
01/01/2014	41640	=DATEVALUE("01/01/2014")

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 | fx TODAY

TODAY

What does it do?

This function returns today's date in date format

Syntax

=TODAY()

No paramter

No paramter is required for this function

Example

Identify today's date

Result	Formula
3/13/2026	=TODAY()
3/8/2026	=TODAY()-5

Exercise

Calculate number of days between today's date and date in Col A

Col A	Result	Formula
12/31/2015	3725	=TODAY()-A21
1/3/2010	5913	=TODAY()-A22
12/11/2015	3745	=TODAY()-A23
1/1/2014	4454	=TODAY()-A24

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 fx NOW

NOW

What does it do?

This function returns current date and time in excel date and time format

Syntax

=NOW()

No paramter

No paramter is required for this function

Example

Identify today's date and time

Result	Formula
3/13/26 20:45	=NOW()
3/8/26 20:45	=NOW()-5

Exercise

Use NOW() to calculate values in COL A

Col A	Formula	Formula
Current date and time	3/13/26 20:45	=NOW()
Date and Time 7 days ago	3/6/26 20:45	=NOW()-7

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 DAY

DAY

What does it do?

This function returns the day of the month i.e. 1, 2 ..31 in numeric format. Do note that at times, you may have to change the format of the cell to numeric to get the date in numeric format

Syntax

=DAY(SERIAL_NUMBER)

Serial Number

Date from which you want to extract the date part

Example

Identify today's date and time

Col A	Result	Formula
3/13/26 20:45	13	=DAY(A16)
3/8/26 20:45	8	=DAY(A17)

Exercise

Use NOW() to calculate values in COL A and then use DAY function to extract date part

Col A	Results from NOW function	Results from Date function	Formula
Current date and time	3/13/26 20:45	13	=DAY(D22)
Date and Time 7 days ago	3/6/26 20:45	6	=DAY(D23)

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 | fx MONTH

MONTH

What does it do?

This function returns the month in a date in a number format i.e. 1, 2, 3 ..12. Do note that at times, you may have to change the format of the cell to numeric to get the date in numeric format

Syntax

=MONTH(SERIAL_NUMBER)

Serial Number

Date from which you want to extract the month part

Example

Identify today's date and time

Col A	Result	Formula
3/29/17 12:11	3	=MONTH(A16)
2/22/17 12:11	2	=MONTH(A17)

Exercise

Use NOW() to calculate values in COL A and then use MONTH function to extract month

Col A	Results from NOW function	Results from Month function	Formula
Current date and time	3/13/26 20:45	3	=MONTH(D22)
Date and Time 30 days ago	2/11/26 20:45	2	=MONTH(D23)

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 YEAR

YEAR

What does it do?

This function returns the year month in a date in a number format. Do note that at times, you may have to change the format of the cell to numeric to get the date in numeric format

Syntax

=YEAR(SERIAL_NUMBER)

Serial Number

Date from which you want to extract the year part

Example

Identify today's date and time

Col A	Result	Formula
3/29/17 12:11	2017	=YEAR(A16)
2/22/1951	1951	=YEAR(A17)

Exercise 1

Use NOW() to calculate values in COL A and then use YEAR function to extract year

Col A	Results from NOW function	Results from Year function
Current date and time	3/13/26 20:45	2026
Date and Time 365 days ago	3/13/25 20:45	2025

Exercise 2

Use YEAR, MONTH, DAY and DATE functions to get dates in COL A from Row 30 onwards. Row 29 has been done as an example

Date	Results	Formula
1/15/2016	Use this date to calculate dates from Row 30 onwards	
2/15/2016	2/15/2016	=DATE(YEAR(A28),MONTH(A28)+1,DAY(A28))
3/15/2016	3/15/2016	=DATE(YEAR(A29),MONTH(A29)+1,DAY(A29))
4/15/2016	4/15/2016	=DATE(YEAR(A30),MONTH(A30)+1,DAY(A30))
5/15/2016	5/15/2016	=DATE(YEAR(A31),MONTH(A31)+1,DAY(A31))
6/15/2016	6/15/2016	=DATE(YEAR(A32),MONTH(A32)+1,DAY(A32))
7/15/2016	7/15/2016	=DATE(YEAR(A33),MONTH(A33)+1,DAY(A33))
8/20/2017	8/15/2016	=DATE(YEAR(A34),MONTH(A34)+1,DAY(A34))

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 WEEKDAY

WEEKDAY

What does it do?

This function returns a number between 1 and 7 depending upon the day of the week. Default return is - Sunday would be 1, Monday 2...so on till Saturday which will be 7. Do note that at times, you may have to change the format of the cell to numeric to get the date in numeric format

Syntax

=WEEKDAY(SERIAL_NUMBER, [return_type])

Serial Number

Date from which you want to extract the year part

return_type

Default settings i.e. if you do not mention this, Sunday would be 1, Monday 2 and so on. If you select it as 2, Monday will be given 1, Tuesday 2 and so on.

Example

Identify today's weekday number basis different return_type value

Col A	Result	Formula
3/13/2026	6	=WEEKDAY(A20)
3/13/2026	5	=WEEKDAY(A21,2)
3/13/2026	4	=WEEKDAY(A22,3)
3/13/2026	2	=WEEKDAY(A23,14)

Exercise 1

Use WEEKDAY() to calculate weekdays of dates mentioned in 'Date' column using 'Return Value'

Dates	Return Value	Results	Formula
1/1/2017	1	1	=WEEKDAY(A28, 1)
1/1/2017	2	7	=WEEKDAY(A29,2)
1/1/2017	3	6	=WEEKDAY(A30,3)

Exercise 2

If any of the dates in the column below falls on Tuesday, then change that date to Friday. Use Weekday fucntion to check weekdays with return value 1. You will have to use IF function. 1st has been done as an example

Dates	Weekday	New Date	Formula
3/28/2017	3	3/24/2017	=IF(B36=3,A36-4,A36)
3/29/2017	4	3/29/2017	=IF(B37=3,A37-4,A37)
3/21/2017	3	3/17/2017	=IF(B38=3,A38-4,A38)
1/1/2016	6	1/1/2016	=IF(B39=3,A39-4,A39)
1/1/2017	1	1/1/2017	=IF(B40=3,A40-4,A40)
1/1/2015	5	1/1/2015	=IF(B41=3,A41-4,A41)
1/1/2014	4	1/1/2014	=IF(B42=3,A42-4,A42)
1/1/2013	3	12/28/2012	=IF(B43=3,A43-4,A43)

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A1:B1 MINUTE

MINUTE

What does it do?

This function returns the minutes between 0 and 59

Syntax

=MINUTE(SERIAL_NUMBER)

Serial Number

Date or time from which you want to extract minutes

Example

Identify today's weekday number basis different return_type value

Col A	Result	Formula
3/13/26 20:45	45	=MINUTE(A15)
10:15 AM	15	=MINUTE(A16)
10:20 PM	20	=MINUTE(A17)

Exercise

Use MINUTE() to find out minutes from Col A

Col A	Results	Formula
1/31/17 0:50	50	=MINUTE(A22)
1/1/17 0:01	1	=MINUTE(A23)
12:45 AM	45	=MINUTE(A24)

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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G1

VLOOKUP

What does it do?

This function is used to lookup a particular value in a table and then return data from a specific column corresponding to that value

Syntax

=VLOOKUP(LookupValue, Table_Array, Column_Index_Num, [Range_Lookup])

LookupValue

The value which needs to be looked for in a particular table

Table_Array

The table in which we are checking for this value to be present and from which we want to retrieve data. By pressing F2 key when you have selected your table, you fix the range. So when you copy formula from one cell to another, the range does not update automatically

Column_Index_Num

If the value is present in the table, then which column consists of the value we want the function to return

Range_Lookup

It is optional - it has 2 values - TRUE for approximate match and FALSE (which is default value) for exact match of the value we are searching for in a table

Example 1

Below in the **Lookup** table, we have cost for different parts for different car makers. Use VLOOKUP() to get costs infront of items in the second table

Lookup Table	Mercedes	BMW	Audi
GearBox	$2,500	$2,000	$3,000
Engine	$5,000	$4,999	$5,200
Steering	$1,250	$1,302	$1,100
Ignition	$250	$300	$200
CYHead	$300	$290	$310

Maker	Spare	Cost	Formula
Mercedes	Ignition	£250	=VLOOKUP(B31,$A$24:$B$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B32,$A$24:$C$28,3,FALSE)
Audi	Engine	£5,200	=VLOOKUP(B33,$A$24:$D$28,4,FALSE)
Audi	Steering	£1,100	=VLOOKUP(B34,$A$24:$D$28,4,FALSE)
Mercedes	Ignition	£250	=VLOOKUP(B35,$A$24:$B$28,2,FALSE)
Mercedes	CYHead	£300	=VLOOKUP(B36,$A$24:$B$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B37,$A$24:$C$28,3,FALSE)
BMW	Engine	£4,999	=VLOOKUP(B38,$A$24:$C$28,3,FALSE)

1) The first parameter relates to spare part for which we want to get the costs
2) The second part refers to the data range - so Mercedes spare part information is available in second column ofthe 'LookUp table' - so that is why you select only first 2 columns - 1st one consists of the spare part name which we are searching for and the second one consists of its cost. For BMW that information is available in 3rd column, thus our data range consists of 3 columns for the second paramter
3) The 3rd parameter refers to from which column we want to return the value, for Mercedes that would be 2nd column of the selected range, for BMW it would be 3rd and for Audi it would be 4th
4) FALSE refers to us wanting an exact match and not an approximate one

Example 2

Below are 3 tables, the first one consists of costs of 4 Items, the second one consists of its sales price. Use VLOOKUP function to put the cost and sales price in 3rd table and then use Units and the difference between Sales Price and Cost to calculate profit or loss

Table 1	
Item Name	Cost
Pen	$5.10
Pencil	$1.10
Notepad	$1.50
Ink	$2.50

Table 2	
Item Name	Sales Price
Pen	$7.50
Pencil	$1.15
Notepad	$2.50
Ink	$3.75

Table 3						
Item Name	Units	Cost	Sales Price	Total Profit	Formula for Cost	Formula for Sales Price
Pen	500	$5.10	$7.50	$1,200.00	=VLOOKUP(A54,$A$46:$B$49,2,FALSE)	=VLOOKUP(A54,$D$46:$E$49,2,FALSE)
Pencil	100	$1.10	$1.15	$5.00	=VLOOKUP(A55,$A$46:$B$49,2,FALSE)	=VLOOKUP(A55,$D$46:$E$49,2,FALSE)
Notepad	250	$1.50	$2.50	$250.00	=VLOOKUP(A56,$A$46:$B$49,2,FALSE)	=VLOOKUP(A56,$D$46:$E$49,2,FALSE)

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A1:B1 fx HOUR

HOUR

What does it do?

This function returns the hour from 0 to 23

Syntax

=HOUR(SERIAL_NUMBER)

Serial Number

Date or time from which you want to extract the hour

Example

Identify today's weekday number basis different return_type value

Col A	Result	Formula
3/13/26 20:45	20	=HOUR(A15)
10:15 AM	10	=HOUR(A16)
11:59 PM	23	=HOUR(A17)

Exercise

Use HOUR() to find out hour from Col A

Col A	Results	Formula
1/31/17 0:50	0	=HOUR(A22)
1/1/17 23:01	23	=HOUR(A23)
5:45 AM	5	=HOUR(A24)

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:C1 HLOOKUP

HLOOKUP

What does it do?

This function is similar to VLOOKUP. In a VLOOKUP, we look at columns, whereas in HLOOKUP, we look at rows. The HLOOKUP function looks for a value in the first row of a table, and returns a value from the same column in that table.

Syntax

=HLOOKUP(LookupValue, Table_Array, Row_Index_Num, [Range_Lookup])

LookupValue

The value which needs to be looked for in a particular table

Table_Array

The table in which we are checking for this value to be present and from which we want to retrieve data. By pressing F2 key when you have selected your table, you fix the range. So when you copy formula from one cell to another, the range does not update automatically

Row_Index_Num

If the value is present in the table, then which row consists of the value we want the function to return

Range_Lookup

It is optional - it has 2 values - TRUE for approximate match (which is default) and FALSE for exact match of the value we are searching for in a table

Example

Below in the first table, we have sales commission for different cars. Use VLOOKUP() to get amount each employee will get based on car they sold

Car Type	Mercedes	BMW	Audi	VW	Ferrari
Commission	$500	$450	$550	$200	$1,000

Sold By	Maker	Commission	Formula
Ben	Mercedes	£500	=HLOOKUP(B29,$B$25:$F$26,2,FALSE)
Ben	BMW	£450	=HLOOKUP(B30,$B$25:$F$26,2,FALSE)
Alex	Audi	£550	=HLOOKUP(B31,$B$25:$F$26,2,FALSE)
Gary	Audi	£550	=HLOOKUP(B32,$B$25:$F$26,2,FALSE)
Helen	Ferrari	£1,000	=HLOOKUP(B33,$B$25:$F$26,2,FALSE)
Helen	Ferrari	£1,000	=HLOOKUP(B34,$B$25:$F$26,2,FALSE)
Ken	BMW	£450	=HLOOKUP(B35,$B$25:$F$26,2,FALSE)
Arnold	Mercedes	£500	=HLOOKUP(B36,$B$25:$F$26,2,FALSE)

1) The first parameter relates to car for which we want to get the commission amount
2) The second part refers to the data range - so car names are in the 1st row of table 1 and the commission information is in 2nd row of table 1
3) The 3rd parameter refers to from which row we want to return the value - in this case it is the 2nd row which consists of the commission amount
4) FALSE refers to us wanting an exact match and not an approximate one

Exercise

Below in the **Lookup** table, we have cost for different parts for different car makers. Use HLOOKUP() to get costs of items in the second table. First one has been done as an example

Lookup Table	Mercedes	BMW	Audi
GearBox	$2,500	$2,000	$3,000
Engine	$5,000	$4,999	$5,200
Steering	$1,250	$1,302	$1,100
Ignition	$250	$300	$200
CYHead	$300	$290	$310

Maker	Mercedes	BMW	Audi	Audi	Mercedes	Mercedes	BMW	BMW	formula
Spare	CYHead	GearBox	Gearbox	Steering	Ignition	Engine	GearBox	Engine	=HLOOKUP(C50, $B$42:$E$47, 2, FALSE)
Costs	£300	£2,000	$3,000	$1,100	$250	$5,000	$2,000	$4,999	=HLOOKUP(D50, $B$42:$E$47, 2, FALSE)
	=HLOOKUP(B50,$B$42:$B$47,6,FALSE)								=HLOOKUP(E50, $B$42:$E$47, 4, FALSE)
									=HLOOKUP(F50, $B$42:$E$47, 5, FALSE)
									=HLOOKUP(G50, $B$42:$E$47, 3, FALSE)
									=HLOOKUP(H50, $B$42:$E$47, 2, FALSE)

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 MATCH

MATCH

What does it do?

This function is used to locate the position of a lookup value in a row, column, or table.

Syntax

=MATCH (Lookupvalue, Lookup_Array, [Match_type])

LookupValue

The value which needs to be looked for in a particular column, row or table

Lookup_Array

The column, table or table in which we are checking for this value to be present

Match_Type

It is optional - it has 3 values --> -1 gives the smallest value position that is greater than or equal to value we are searching, 0 gives the value for the exact search and 1 gives the largest value that is leass than or equal to the value we are searching for. Default is 1

Example 1

Below in the **Lookup** table, we have cost for different parts for different car makers. Use VLOOKUP() to get costs infront of items in the second table

Lookup Table	Mercedes	BMW	Audi
GearBox	$2,500	$2,000	$3,000
Engine	$5,000	$4,999	$5,200
Steering	$1,250	$1,302	$1,100
Ignition	$250	$300	$200
CYHead	$300	$290	$310

Maker	Spare	Cost	Formula
Mercedes	Ignition	£250	=VLOOKUP(B31,$A$24:$B$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B32,$A$24:$C$28,3,FALSE)
Audi	Engine	£5,200	=VLOOKUP(B33,$A$24:$D$28,4,FALSE)
Audi	Steering	£1,100	=VLOOKUP(B34,$A$24:$D$28,4,FALSE)
Mercedes	Ignition	£250	=VLOOKUP(B35,$A$24:$B$28,2,FALSE)
Mercedes	CYHead	£300	=VLOOKUP(B36,$A$24:$B$28,2,FALSE)
BMW	GearBox	£2,000	=VLOOKUP(B37,$A$24:$C$28,3,FALSE)
BMW	Engine	£4,999	=VLOOKUP(B38,$A$24:$C$28,3,FALSE)

1) The first parameter relates to spare part for which we want to get the costs
2) The second part refers to the data range - so Mercedes spare part information is available in second column ofthe 'LookUp table' - so that is why you select only first 2 columns - 1st one consists of the spare part name which we are searching for and the second one consists of its cost. For BMW that information is available in 3rd column, thus our data range consists of 3 columns for the second paramter
3) The 3rd parameter refers to from which column we want to return the value, for Mercedes that would be 2nd column of the selected range, for BMW it would be 3rd and for Audi it would be 4th
4) FALSE refers to us wanting an exact match and not an approximate one

Example 2

Below are 3 tables, the first one consists of costs of 4 Items, the second one consists of its sales price. Use VLOOKUP function to put the cost and sales price in 3rd table and then use Units and the difference between Sales Price and Cost to calculate profit or loss

Table 1	
Item Name	Cost
Pen	$5.10
Pencil	$1.10
Notepad	$1.50
Ink	$2.50

Table 2	
Item Name	Sales Price
Pen	$7.50
Pencil	$1.15
Notepad	$2.50
Ink	$3.75

Table 3						
Item Name	Units	Cost	Sales Price	Total Profit	Formula for Cost	Formula for Sales Price
Pen	500	$5.10	$7.50	$1,200.00	=VLOOKUP(A54,$A$46:$B$49,2,FALSE)	=VLOOKUP(A54,$D$46:$E$49,2,FALSE)
Pencil	100	$1.10	$1.15	$5.00	=VLOOKUP(A55,$A$46:$B$49,2,FALSE)	=VLOOKUP(A55,$D$46:$E$49,2,FALSE)
Notepad	250	$1.50	$2.50	$250.00	=VLOOKUP(A56,$A$46:$B$49,2,FALSE)	=VLOOKUP(A56,$D$46:$E$49,2,FALSE)
Ink	100	$2.50	$3.75	$125.00	=VLOOKUP(A57,$A$46:$B$49,2,FALSE)	=VLOOKUP(A57,$D$46:$E$49,2,FALSE)
Eraser	990	#N/A	#N/A		=VLOOKUP(A58,$A$46:$B$49,2,FALSE)	=VLOOKUP(A58,$D$46:$E$49,2,FALSE)

Total Profit = (Sales Price-Cost)*Units

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

File Edit View Insert Format Data Tools Help

Menus 100% View only

A1:B1 SUM

SUM

What does it do?

This function returns the sum of the cells or range selected

Syntax

=SUM(Number 1, Number 2...)

Number 1

The first value to sum

Number 2 and so on

The second (and so on..) value to sum

Example 1

Use SUM function to get the sum of cells

Col A	Col B	Col C	Result	Formula
5	12	17	34	=SUM(A18:C18)
10	18	23	51	=SUM(A19,B19,C19)
20	100	999	1119	=A20+B20+C20

Example 2

Use SUM function to get the sum of cells

Row A	5	9	12	21
Row B	1	56	345	23
Row C	2	6	23	34
Row D	4	12	547	12
Total	12	83	927	90
Formula	=SUM(B24:B27)	=C24+C25+C26+C27	=SUM(D24,D25,D26,D27)	=SUM(E24:E25,E26:E27)

Example 3

Get the SUM of all cells below

Values to be summed			Total	Formula
5		3	16	=SUM(A34,A36,C34,C36,B35:B36)
	1			
1	2	4		

Note

If you want to sum the entire column I suppose, all you need to do is =SUM(I:I), it will select the entire column I and give you the sum of the numbers in it. Similarly, you can select the entire row as well. If you write, =SUM(41:42), It will return the sum of all numbers in Rows 41 and 42.

Exercise 1

Use SUM function to calculate sum of the following values

Values to be summed			Result	Formula
1			801	=SUM(A45,A46,B47,C46)
100		500		
	200			

Exercise 2

Use IF and SUM function to calculate sum of the following values if conditions meet. If the conditions do not meet, then return value 0. First has been done as an example

Condition to	Col B	Col C	Col D	Col E	Result	
Col B > 1	5	1	2	4	12	**=IF(B54>1,SUM(B54:E54),0)**

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A1:B1 SUMIF

SUMIF

What does it do?

This function returns the sum of the cells which meet a given criteria

Syntax

=SUMIF(Range, Criteria, [sum_range])

Range

Range of cells which we need the criteria to evaluate

Criteria

This refers to the condition which we want to check, it can be text, number, dates

Sum Range

This is optional, if not present, excel will consider numbers in the 'Range'. Otherwise, it specifies the cells which need to be added if the condition holds

Example 1

Use SUMIF function to calculate salary of all employees in Analytics and then for Sales

Employee ID	Department	Designation	Date of Joining	Salary
E001	HR	AVP	1/1/2017	$40,000
E002	Analytics	AVP	12/1/2016	$75,000
E003	Sales	AVP	12/1/2016	$30,000
E004	Analytics	AVP	11/1/2016	$80,000
E005	HR	Analyst	10/1/2016	$35,000
E006	Sales	Analyst	11/1/2016	$25,000
E007	Analytics	Analyst	10/1/2016	$60,000
E008	Sales	Analyst	11/1/2016	$20,000
E009	IT	Analyst	10/1/2016	$40,000
E010	IT	Analyst	10/1/2016	$38,000

	Total	Formula
Salary for Analytics Group	$215,000	=SUMIF(B22:B31,"Analytics",E22:E31)
Salary for Analytics Group	$78,000	=SUMIF(B22:B31,"IT",E22:E31)

Example 2

Use SUMIF function to calculate salary of all employees who joined before 1st Dec 2016

Employee ID	Department	Designation	Date of Joining	Salary
E001	HR	AVP	1/1/2017	$40,000
E002	Analytics	AVP	12/1/2016	$75,000
E003	Sales	AVP	12/1/2016	$30,000
E004	Analytics	AVP	11/1/2016	$80,000
E005	HR	Analyst	10/1/2016	$35,000
E006	Sales	Analyst	11/1/2016	$25,000
E007	Analytics	Analyst	10/1/2016	$60,000
E008	Sales	Analyst	11/1/2016	$20,000
E009	IT	Analyst	10/1/2016	$40,000
E010	IT	Analyst	10/1/2016	$38,000

	Total	Formula
Salary for employees who joined before 1st Dec'16	$298,000	=SUMIF(D41:D50,"<01-Dec-2016",E41:E50)

Exercise 1

MATH - SUMIF | MATH - SUMPRODUCT | MATH - ROUND | MATH - EVEN | MATH - MOD | STATISTICAL - MIN | STATISTICAL

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A1:C1 SUMPRODUCT

SUMPRODUCT

What does it do?

This function returns the sum after multiplying numbers in an array. It can also be used to count cells based on criteria provided

Syntax

=SUMPRODUCT(Array1, [Array2], [Array3],...and so on)

Array1

Range of cells which first need to be multiplied and then summed

Array2

It is optional - the second array or range to multiply and then summed

Example 1

Use SUMPRODUCT function to calculate total cost faced by the superstore

Items	Perishable?	Cost	No. of Units
I001	NO	£1.50	100
I002	YES	£1.99	150
I003	YES	£2.05	200
I004	NO	£3.50	50
I005	YES	£4.90	400
I006	NO	£5.00	1,000

	Total	Formula
Total Cost	$7,994	=SUMPRODUCT(C18:C23,D18:D23)

Example 2

Use SUMPRODUCT function to count total number of Perishable items

Items	Perishable?	Total Cost
I001	NO	£150
I002	YES	£299
I003	YES	£410
I004	NO	£175
I005	YES	£1,960
I006	NO	£5,000

	Total	Formula
Total Items	3	=SUMPRODUCT(--(B32:B37="YES"))
Cost for Perishable Items	$2,669	=SUMPRODUCT(--(B32:B37="YES"),C32:C37)

Exercise 1

Use SUMPRODUCT function to calculate total cost faced by the superstore

Items	Perishable?	Sales	No. of Units
I001	NO	£3.10	500
I002	YES	£2.50	150
I003	YES	£10.00	100
I004	NO	£1.00	500
I005	YES	£4.00	200
I006	NO	£2.00	2,000

	Total	Formula
Total Sales	$8,225	=SUMPRODUCT(C47:C52,D47:D52)

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A1:B1 ROUND

ROUND

What does it do?

This function rounds the numbers to specified digits.

Syntax

=ROUND(Number,num_digits)

Number

Number which you want to round up

num_digits

Number of decimal places to which you want it to round up

Example

Use ROUND function reduce decimal places of the numbers

Numbers	Rounded	Formula
45.3976106	45.40	=ROUND(A18,2)
50.6546265	50.70	=ROUND(A19,1)
85.4139093	85.414	=ROUND(A20,3)
33.5893471	34	=ROUND(A21,0)

Exercise

Use ROUND function reduce decimal places of the numbers

Items	Result	Formula
81.373	81.37	=ROUND(A26,2)
7.9597	8.00	=ROUND(A27,2)
521.2578	521.26	=ROUND(A28,2)
4.699	4.70	=ROUND(A29,2)

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A1:B1 | fx EVEN

EVEN

What does it do?

This function rounds a positive number up and negative number down to the nearest even integer

Syntax

=EVEN(Number/Cell)

Text

Cell which you want to convert to the nearest even integer

Example

Use EVEN function on the following examples

Numbers	Rounded	Formula
45.3976106	46	=EVEN(A15)
-50.6546265	-52	=EVEN(A16)
85.4139093	86	=EVEN(A17)
31.5893471	32	=EVEN(A18)

Exercise

Use EVEN function and round up numbers to nearest even number

Items	Result	Formula
1111.111	1,112.00	=EVEN(A23)
95.3	96.00	=EVEN(A24)
51.5	52.00	=EVEN(A25)
-4.699	-6.00	=EVEN(A26)

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A1:B1 fx MOD

MOD

What does it do?

This function returns the remainder after a number is divided by a divisor

Syntax

=MOD(number,divisor)

Number

This is the number which you want to divide

Divisor

The number to divide with

Example

Use MOD function on the following examples

Number	Divisor	Remainder	Formula
3	5	3	=MOD(A18,B18)
19	3	1	=MOD(A19,B19)
8	2	0	=MOD(A20,B20)
-8.5	3	0.5	=MOD(A21,B21)

Exercise

Use MOD function to find the remainders

Number 1	Number 2	Result	Formula
5	6	5	=MOD(A26,B26)
9	2	1	=MOD(A27,B27)
8	5	3	=MOD(A28,B28)
-4.699	-1	-0.699	=MOD(A29,B29)

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A1:B1 fx MIN

MIN

What does it do?

This function returns the smallest number in a given set of values. It ignores logical values and text

Syntax

=MIN(Number1,Number2 and so on..)

Number1,Number2 and so on..

List of values from which you want to find the mimimum value

Example

Use MIN function to find the smallest value of the 3 numbers present for each row

Number1	Number2	Number3	Result	Formula
3	5	3	3	=MIN(A15:C15)
19	a	1	1	=MIN(A16:C16)
8	b	c	8	=MIN(A17:C17)
-8.5	3	0.5	-8.5	=MIN(A18:C18)

Exercise

Use MIN and IF function on the following and if the value is less than 5, then flag it as 'Less than 5' else 'Greater than equal to 5'

Number1	Number2	Number3	Result	Formula
3	5	3	3	=MIN(A23:C23)
19	a	1	1	=MIN(A24:C24)
8	b	c	8	=MIN(A25:C25)
-8.5	3	0.5	-8.5	=MIN(A26:C26)

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O30

MAX

What does it do?

This function returns the largest number in a given set of values. It ignores logical values and text

Syntax

=MAX(Number1,Number2 and so on..)

Number1,Number2 and so on..

List of values from which you want to find the maximum value

Example

Use MAX function to find the smallest value of the 3 numbers present for each row

Number1	Number2	Number3	Result	Formula
3	5	3	5	=MAX(A15:C15)
19	a	1	19	=MAX(A16:C16)
8	b	c	8	=MAX(A17:C17)
-8.5	3	0.5	3	=MAX(A18:C18)

Exercise

Use MAX and IF function on the following and if the value is less than 5, then flag it as 'Less than 5' else 'Greater than equal to 5'

Number1	Number2	Number3	Result	Formula
3	5	3	5	=MAX(A23:C23)
19	a	1	19	=MAX(A24:C24)
8	b	c	8	=MAX(A25:C25)
-8.5	3	0.5	3	=MAX(A26:C26)

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A1:B1 AVERAGE

AVERAGE

What does it do?

This function returns the average value of the given numbers - It ignores text values and blank cells

Syntax

=AVERAGE(Number1,Number2 and so on..)

Number1,Number2 and so on..

List of values from which you want to find the average value

Example

Use AVERAGE function to find the smallest value of the 3 numbers present for each row

Number1	Number2	Number3	Result	Formula
3		5	4	=MAX(A15:C15)
19	a	1	10	=MAX(A16:C16)
8	b	c	8	=MAX(A17:C17)
-8.5	3	0.5	-1.666666667	=MAX(A18:C18)

Exercise

Calculate AVERAGE for the following numbers

Number1	Number2	Number3	Result	Formula
3	5	3	5	=MAX(A23:C23)
19	a	1	19	=MAX(A24:C24)
8	b	c	8	=MAX(A25:C25)
1.5		4.5	4.5	=MAX(A26:C26)

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A1:B1 | fx SMALL

SMALL

What does it do?

This function returns the nth smallest number from the given set of values

Syntax

=SMALL(Array, nth small number from the list)

Array

These are list of numbers from which nth smallest number needs to be found out

nth smallest number

The nth smallest number which needs to be found out like 5th smallest number etc

Example

Use SMALL function to find the nth smallest value from the set of values in the range A18:C21

Array/List of Numbers			nth smallest	Result	Formula
3	1	5	1	-8.5	=SMALL(A18:C21,D18)
19	50	1	5	3	=SMALL(A18:C21,D19)
8	12	20	4	1	=SMALL(A18:C21,D20)
-8.5	3	0.5	12	50	=SMALL(A18:C21,D21)

Exercise

Calculate nth smallest number for the following numbers

Array/List of Numbers			nth smallest	Result	Formula
3	5	3	1	1	=SMALL(A26:C29,D26)
19	2	1	2	1	=SMALL(A26:C29,D26)
8	5	3	3	3	=SMALL(A27:C30,D27)
1.5	1	4.5	4	#NUM!	=SMALL(A28:C31,D28)

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A1:B1 LARGE

LARGE

What does it do?

This function returns the nth largest number from the given set of values

Syntax

=LARGE(Array, nth largest number from the list)

Array

These are list of numbers from which nth largest number needs to be found out

nth largest number

The nth largest number which needs to be found out like 5th largest number etc

Example

Use SMALL function to find the nth smallest value from the set of values in the range A18:C21

Array/List of Numbers			nth largest	Result	Formula
3	1	5	1	50	=LARGE(A18:C21,D18)
19	50	1	5	8	=LARGE(A18:C21,D19)
8	12	20	4	12	=LARGE(A18:C21,D20)
-8.5	3	0.5	12	-8.5	=LARGE(A18:C21,D21)

Exercise

Calculate nth largest number for the following numbers

Array/List of Numbers			nth largest	Result	Formula
3	5	3	1	5	=LARGE(A26:C26, D26)
19	2	1	2	2	=LARGE(A27:C27, D27)
8	5	3	3	3	=LARGE(A28:C28, D28)
1.5	1	4.5	4	#NUM!	=LARGE(A29:C29, D29)

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 COUNT

COUNT

What does it do?

This function returns the count of number of cells which consist of numbers

Syntax

=COUNT([value1],[value[2], and so on..)

value1

First cell in the range

value 2 and so on

Second, third and so on cells in the range

Example

Use COUNT function to find the number of numeric values

List of Numbers			Result	Formula
3	1	5	3	=COUNT(A18:C18)
19	A	1	2	=COUNT(A19:C19)
8		20	2	=COUNT(A20:C20)
-8.5		B	1	=COUNT(A21:C21)

Exercise 1

Use COUNT function to find the number of numeric values

List of Numbers			Result	Formula
3		3	2	=COUNT(A26:C26)
b	2	1	2	=COUNT(A27:C27)
c	5	3	2	=COUNT(A28:C28)
1.5	1	4.5	3	=COUNT(A29:C29)

Exercise 2

Use IF function to print 'Y' if count of numbers is >=2 else print 'N'

List of Numbers			Result	Formula
3		3	Y	=IF(COUNT(A34:C34)>=2,"Y","N")
b	2	1	Y	=IF(COUNT(A35:C35)>=2,"Y","N")
c	5	3	Y	=IF(COUNT(A36:C36)>=2,"Y","N")
1.5		D	N	=IF(COUNT(A37:C37)>=2,"Y","N")

Practice Task 3._Gilhang_Serad - Advanced Excel Functions .XLSX

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A1:B1 COUNTIF

COUNTIF

What does it do?

This function returns the count of number of cells which consist of numbers and meet a given condition

Syntax

=COUNTIF(Range,Criteria)

Range

Range of cells which you want to count

Criteria

Condition which you want to check and then count the cells on the basis of it

Example

Use COUNTIF function on Col A, B, C and D to find the number of numeric values, based on values being greater or less as specified in Col E

Col A	Col B	Col C	Col D	Col E	Result	Formula
3	1	5	3	Values > 1	3	=COUNTIF(A18:D18,">1")
19	A		0	Values = 0	1	=COUNTIF(A19:D19,"=0")
8	D	20	2	Values >= 2	3	=COUNTIF(A20:D20,">=2")
-8.5	0	B	-1	Values < 0	2	=COUNTIF(A21:D21,"<0")

Exercise

Use COUNTIF function on Col A, B, C and D to find the number of numeric values, based on values being greater or less as specified in Col E

Col A	Col B	Col C	Col D	Col E	Result	Formula
1	-1	5	3	Values > 2	2	=COUNTIF(A26:D26,">2")
19	A		0	Values >= 0	2	=COUNTIF(A27:D27,">=0")
8	-50	20	2	Values >= 20	1	=COUNTIF(A28:D28,">=20")
-8.5	0		-1	Values <= 0	3	=COUNTIF(A29:D29,"<=0")